Human Computer Interaction - Group A

AKSHIT KOTNALA - 4302

Army Institute of Technology, Pune SPPU University

July 28, 2025

Contents

Identifying and observing bad designs

Bad Designs

1. Bad design refers to the creation of products, systems, interfaces, or experiences that are ineffective, inefficient, confusing, or aesthetically displeasing.
2. When a design doesn't take into account the needs, preferences, or behaviors of the intended users, it's likely to be ineffective.
3. Design that fails to serve its intended purpose or function is considered bad.
4. Bad Design, often lacks a clear purpose and can be difficult to understand.
5. Bad Design is visually cluttered and unbalanced, making it difficult for the viewer to focus on the important information.

Bad Design - Example 1

- Ceiling with exposed wiring, and a persistent leak during rainy days presents a visually unpleasant and hazardous living space.

Bad Design - Example 2

- The poorly maintained AIT hostel window features rusted bars that impede its proper opening

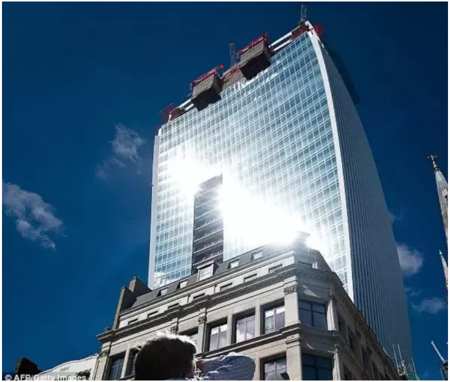
Bad Design - Example 3

- Poorly designed disabled shower room door, hindering independent bathing for individuals with mobility impairments.

Bad Design - Example 4

- The chair's single, fixed writing surface may not be ergonomically adjustable for users of varying heights and postures and mainly with left hand writters.

Bad Design - Example 5



- Building's reflective facade concentrates sunlight into a powerful beam, causing damage to nearby property and posing a safety hazard to pedestrians.

Bad Design - Example 6

- No gaps are provided for USB cable to get plugged into a Mini DisplayPort adapter, indicating a potential misuse of the adapter and a possible risk of damaging the laptop's port.

Good Design

Good Design

1. **The design should be visually pleasing and attractive to the target audience..**
2. **It should be easy to understand and use, with intuitive navigation and clear instructions.**
3. Use size, color, and placement to create a visual hierarchy that guides the user's attention.

Good design - Example 1

- The washing product boasts a sleek, professional design, ergonomic grip, and an innovative lid doubling as a single-use measurement marker.

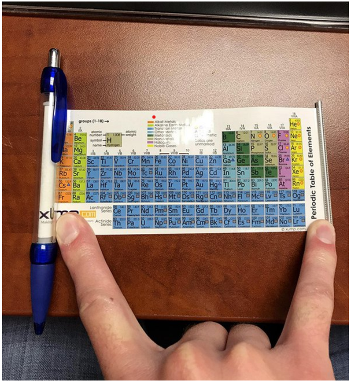
Good design - Example 2

- The squat lunge machine f with a sturdy frame and ergonomic handles, emphasizing functionality and visual appeal.

Good Design - Example 3

- Yellow trash can , standing on a cobblestone sidewalk, which provides cyclist to throw their garbage diractly without waiting to stand up.

Good Design - Example 4



- Compact, portable periodic table pen which helps in making it easy to reference and study.

Good Design - Example 5

- A simple compact design of sitting areas to utilize the minimum space with maximum efficiency.

Good Design - Example 6

- Creatively designed pizza menu where each page is shaped like a pizza, allowing customers to visualize the actual pizza they are ordering.

Good Design - Example 7

- Shopping cart with a built-in calculator, allowing shoppers to estimate their total purchases while shopping.

Feedbacks

Feedbacks

1. Feedback refers to the information or response that a system provides to a user's actions, inputs, or interactions.
2. Feedback is a crucial element in designing user interfaces and interactions to ensure that users understand the outcome of their actions and can effectively use digital systems.
3. It helps users navigate interfaces, make informed decisions, and achieve their goals.

Feedback - Example 1

- Thermostat that displays options, clarity, and energy efficiency features.

Feedback - Example 2

- Washing machine's control panel.

Feedback - Example 3

- Smoke detector is detecting smoke and is likely functioning correctly.

Feedback - Example 4

- Keyboard's design and lighting provide feedback for correct usage.

Feedback - Example 5

- Kettle's operation and its current status in its display screen.

Constraints

Constraints

1. **Constraints refer to limitations or restrictions that are intentionally designed into user interfaces, systems, or interactions to guide user behavior, prevent errors, and ensure appropriate interactions.**
2. **Constraints are used to shape user actions and decisions in a way that aligns with the intended design and functionality of a system.**
3. **They help users understand how to interact with a system more effectively and reduce the likelihood of mistakes.**

Constraints - Example 1

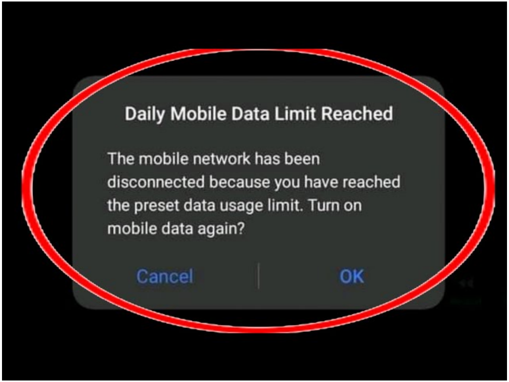
Rules at an airportcenter

- Constraints are related to the size, weight (10 kg maximum), and the number of bags (max 1 bag) that passengers can bring onboard.

Constraints - Example 2

- Speed limit sign on a highway
- Constraint here is the speed regulation

Constraints - Example 3

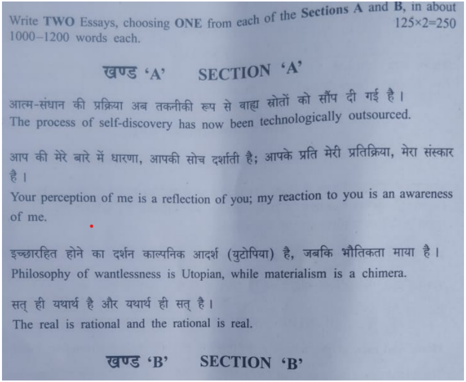


- Notification message on a mobile device.
- The constraint here is the limit on daily mobile data usage, beyond which the user is disconnected.

Constraints - Example 4

- Blue sign which displays opening hours and a maximum capacity.
- Constraints: Time-bound: The schedule specifies distinct morning and afternoon time slots for operation.

Constraints - Example 5



Write TWO Essays, choosing ONE from each of the Sections A and B, in about 1000–1200 words each. 125×2=250

खण्ड 'A' SECTION 'A'

आत्म-संधान की प्रक्रिया अब तकनीकी रूप से बाह्य स्रोतों को सौंप दी गई है।
The process of self-discovery has now been technologically outsourced.

आप की मेरे बारे में धारणा, आपकी सोच दर्शाती है; आपके प्रति मेरी प्रतिक्रिया, मेरा संस्कार है।
Your perception of me is a reflection of you; my reaction to you is an awareness of me.

इच्छारहित होने का दर्शन काल्पनिक आदर्श (युटोपिया) है, जबकि भौतिकता माया है।
Philosophy of wantlessness is Utopian, while materialism is a chimera.

सत् ही यथार्थ है और यथार्थ ही सत् है।
The real is rational and the rational is real.

खण्ड 'B' SECTION 'B'

- essay prompt in both Hindi and English
- The constraint here is the requirement to adhere to the given themes and stay within the specified word count.

Jugaad

Jugaad

1. **Jugaad refers to creative and resourceful approaches to solving design and interaction challenges**
2. **Jugaad involves finding innovative and unconventional solutions using the available resources and technologies.**

Jugaad - Example 1

- To close the door without latching, extra cushion is added.

Jugaad - Example 2

- Bicycle is secured with a pair of humble slippers!

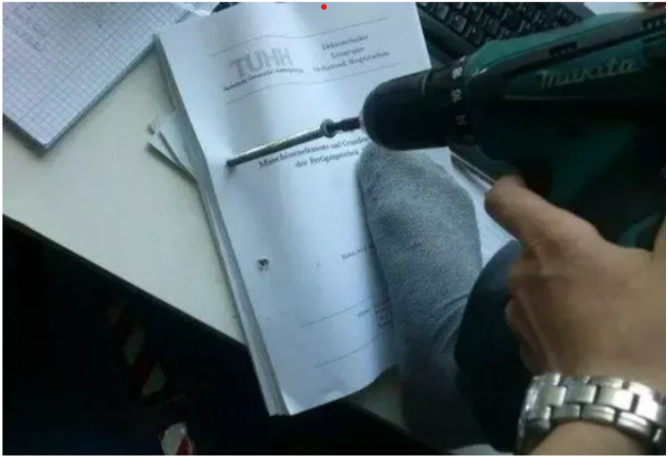
Jugaad - Example 3

- Keeping a cold drink chilled without a refrigerator.

Jugaad - Example 4

- Makeshift mobile phone charging station, set up using a car battery, wires, and various chargers.

Jugaad - Example 5



- Using a drill machine to a screw in place through a piece of paper.

Thank You!

Any questions?